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Explainable Artificial Intelligence

**Enhancing Interpretability of Deep Neural Networks
Using Visual Analytics**



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Abstract

In this work, we discuss various aspects related to artificial intelligence (AI) systems, the need for understanding AI, explainable AI techniques, visual analytics approaches, and multi-modal contrastive learning models. Despite the power of deep neural networks, they are often considered "black box models" due to their complex inner workings. The text stresses the importance of understanding AI and mentions how explanations can enhance transparency, improve model performance, and calibrate user trust. There are various kinds of explainable AI techniques that can be used for interpreting deep learning models. However, interactive visualization tools are needed for effectively communicating complex data. Visual analytics techniques are indispensable to gain a dynamic view of deep neural networks, enabling users to explore explanations for multiple instances and gain insights into high-dimensional data through dimensionality reduction methods like t-SNE.

The recent emergence of multi-modal contrastive learning models opens a whole new area of investigation in the field of deep neural networks. Multi-modal contrastive learning models bridge the gap between text and images by learning representations that connect the two modalities. We investigated these learned representations, and analyzed the modality gap of multi-modal contrastive models, how to close it, and why closing it is important for downstream tasks. We developed an interactive visual analytics tool for investigating multi-modal contrastive models and their embedding spaces to aid the analysis. We analyzed a number of models and how they differed with regard to the modality gap. Furthermore, we analyzed various sets and subsets of data and were able to find interesting clustering in the proposed visualizations. By visualizing augmented versions of single images, we were able to analyze, how robust each model is with regard to the augmentation. The tool is an open-source python package and allows model developers to easily probe their own models in a jupyter notebook environment.

Contents

1	Introduction	1
1.1	Artificial Intelligence Systems	1
1.2	The Need to Understand AI	1
1.3	Explainable AI Techniques	2
1.4	Visual Analytics Approaches	3
1.5	Multi-Modal Contrastive Learning Models	3
2	Understanding and Comparing Multi-Modal Models	5
2.1	CLIP and its Modality Gap	5
2.2	Where does the modality gap come from?	6
2.3	What About Other CLIP-Like Models?	9
2.4	Closing the Gap Using a Larger Dataset	13
2.5	Visually Closing the Modality Gap	15
2.6	Summary of Modality Gap Analysis	18
3	Converting the Technique into a Tool	20
3.1	Identifying Data Subsets	20
3.2	Analyze DiffusionDB Dataset	20
3.3	Augmentation Analyses	22
4	Conclusion	26
	References	28

List of Figures

2.1	A sketch of a projected bi-modal embedding space, as originally expected.	5
2.2	Projected image and text embedding space as generated by CLIP [13].	6
2.3	Similarity of image and text embeddings visualized as heatmap.	7
2.4	Schema plot of how to close the modality gap by moving the embeddings along the gap vector.	7
2.5	Loss changes when moving the embeddings of two modalities in CLIP closer together or further apart. The dashed line marks the original distance of the two modalities.	8
2.6	Similarity heatmap and two-dimensional CLIP embeddings.	8
2.7	Similarity heatmap and two-dimensional CLIP embeddings.	9
2.8	Loss changes when moving the embeddings of two modalities in CyCLIP closer together or further apart. The dashed line marks the original distance of the two modalities.	10
2.9	Similarity heatmap and two-dimensional CyCLIP embeddings.	11
2.10	Two-dimensional CLIP embeddings projected with UMAP.	11
2.11	Similarity heatmap and two-dimensional CLOOB embeddings.	13
2.12	Two-dimensional projection of CyCLIP embeddings with the modality gap manually removed.	14
2.13	Two-dimensional projection of CLIP embeddings with the modality gap manually removed.	14
2.14	Two-dimensional projection of CLOOB embeddings with the modality gap inherently removed.	15
2.15	Two-dimensional projection of CLOOB Laion400M embeddings with the modality gap inherently removed.	15
2.16	Two-dimensional projection of CLIP embeddings with the modality gap removed using out-of-sample projection of the text embeddings.	17
2.17	Two-dimensional projection of CLIP embeddings where image and text embeddings were concatenated to one datapoint.	17
2.18	Similarity heatmaps for all CLIP-like models and their closed embeddings.	19
3.1	Interactive tool to analyze multi-modal models "Amumo".	20
3.2	Using Amumo to analyze a data subset of "dog". Reordering the similarity heatmap reveals a cluster that shows hot dogs instead of the animal.	21
3.3	Using Amumo to analyze the DiffusionDB dataset.	21
3.4	Using Amumo to analyze embeddings of rotated versions of an image.	23
3.5	Using Amumo to analyze embeddings of noisy versions of an image.	24
3.6	Using Amumo to analyze embeddings of blurry versions of an image.	25

1 Introduction

1.1 Artificial Intelligence Systems

Machine learning (ML) models, a subset of artificial intelligence (AI) systems, have seen extensive use and application across a multitude of domains and industries in today's technology-driven world. These AI technologies have made significant strides, enabling automation in critical processes, enhancing decision-making, and even fostering creative content generation. One remarkable application of AI is in the realm of autonomous transportation, exemplified by self-driving cars [4]. These vehicles rely on intricate ML algorithms to navigate and make real-time decisions, demonstrating AI's potential to revolutionize the future of transportation by ensuring safer and more efficient journeys. In the field of healthcare, AI has emerged as a powerful ally in optimizing clinical decisions and diagnoses [16]. AI-driven medical diagnosis systems can analyze vast amounts of patient data, including medical images and patient records, to assist healthcare professionals in making more accurate and timely assessments. This capability has the potential to greatly enhance patient care and reduce the burden on medical practitioners.

Furthermore, AI has unlocked the realm of creative content generation, as exemplified by Text-to-Image Generation [15, 14]. These AI systems possess the ability to convert textual descriptions into visually stunning images, opening up new possibilities in design, advertising, and storytelling. Such AI-driven content generation not only streamlines the creative process but also enables the rapid production of customized visuals tailored to specific needs. In parallel with these domain-specific AI applications, the AI landscape is currently witnessing the emergence of Foundation Models [1]. These Foundation Models, which are deep neural networks (DNNs), represent a groundbreaking development in the AI field. They undergo training on a vast and diverse range of data types, making them highly adaptable and versatile. Importantly, these models can be transferred to address various task domains, showcasing their potential to serve as the building blocks for a wide array of AI applications.

In summary, the pervasive influence of AI, particularly within the realm of ML, extends to numerous domains and applications. From self-driving cars to healthcare diagnostics and creative content generation, AI continues to reshape industries and drive innovation. The advent of Foundation Models further amplifies AI's impact, promising to unlock even more possibilities across diverse task domains, solidifying its status as a transformative force in the modern technological landscape.

1.2 The Need to Understand AI

While it is undeniable that deep neural networks have exhibited remarkable performance in surpassing human capabilities across various tasks, there exists a crucial caveat to their reliability — the potential for unexpected errors and biases in their predictions.

Detecting and comprehending these errors is a task of paramount importance, albeit one marked by its inherent complexity.

Deep neural network models, by their deterministic nature, offer a sense of confidence in their decisions. However, paradoxically, they are often labeled as "black box models" due to the intricate and deeply nested mathematical and statistical computations that underlie their training and inference processes. This complex internal structure poses a significant challenge for human understanding. The inner workings of these models, characterized by high-dimensional weights, activations, and feature maps, remain elusive without the aid of abstraction. The introduction of explanations into the AI landscape plays a pivotal role in addressing this challenge. Explanations serve as the beacon illuminating the murky depths of AI systems, aiding humans in gaining a deeper and more transparent understanding of their operations. This heightened comprehension brings about several notable advantages. For one, machine learning engineers can leverage explanations to refine and enhance model performance. Explanations provide valuable insights into the model's decision-making processes, facilitating debugging and the identification and mitigation of learned biases. Moreover, explanations hold immense promise in human-AI collaborations, particularly in the context of AI-assisted decision-making. Users of AI systems can utilize explanations to calibrate their trust in these systems. Trust calibration becomes a critical factor, as users should place their trust in AI systems only when the predictions are correct. By wielding explanations, users can gauge the reliability of AI-generated recommendations and thus make more informed decisions.

In essence, the integration of explanations into AI systems marks a significant step forward in bridging the gap between the often opaque nature of deep neural networks and the human need for transparency and trust. Through explanations, the path toward improved model performance, bias detection, and trustworthy AI-assisted decision-making becomes not only conceivable but also achievable, fostering more effective and harmonious collaborations between humans and AI systems.

1.3 Explainable AI Techniques

Various visualization techniques can be used for interpreting deep learning models. However, the type of visualization highly depends on the kind of data we want to show, the type of interpretation users want to do, to which target audience users belong, and in which domain and with which types of models they work with.

Global explanation methods, for example, are used to explain the behavior of a machine learning model as a whole. To that group of methods belong techniques that show the effects of input features on output values, or the overall importance of a feature. Also, the visualization of the computational graph of a deep learning network can help model developers understand a network's flow of data. Feature visualization approaches [11] can be used to visualize convolutional neural networks, which are essential for image-based tasks (like image classification). They bring light into the building blocks of convolutional neural networks and help model developers find, which parts of a network are responsible for detecting certain features in an input.

Local explanation methods, on the other hand, help interpret the prediction of single input instances (in the case of image processing one instance would be a single input image). This group of methods contains techniques that calculate attribution

scores, or counterfactual examples. When looking at the domain of image classification again, a lot of approaches calculate important regions of an input image to the predicted result [18]. Those importance scores (also known as saliency maps) are then often shown in the form of heatmaps that are overlaid on top of the target image.

Many XAI techniques generate meaningful, yet static explanations that are often insufficient to understand the processes of deep neural networks. Visual analytics and interactive visualization tools can be used for communicating complex data using visual encodings. Such tools let users explore data and allow them to gain insights and discover anomalies, leading to a better understanding of the underlying system.

1.4 Visual Analytics Approaches

Many techniques exist that produce meaningful insights into deep neural networks. Nevertheless, these techniques usually give a static view of a model or single input instances. Using visual analytics techniques we can create a more dynamic view of the problem at hand. We can, for example, create interactive tools that allow users to not only show the explanation of a single static instance, but that lets them interactively switch between instances, without any programming overhead [20]. Or we can give users an abstracted overview of a model at first and let them explore the model in more detail on demand [8].

Deep neural networks work with high-dimensional data and usually train on large amounts of data. Dimensionality reduction techniques, like t-distributed stochastic neighborhood embedding, can be utilized to visualize such data in a two-dimensional space. The aim of such projection techniques is that similar data points are close to each other in the two-dimensional space. Instances that are supposed to be similar to each other (e.g., because they belong to the same class), but are far away from each other in the projection can be identified and further investigated [12].

With these existing explainability techniques and visual analytics tools as a basis, we plan to develop a prototype of an interactive tool that helps explain state-of-the-art deep learning models. Depending on the application domain and the exact use cases we can adopt existing explainability techniques, or develop new explanation or visualization idioms. The prototype should utilize interactive features allowing users to get an overview of a deep neural network and letting them explore various interesting aspects of the model in more detail.

1.5 Multi-Modal Contrastive Learning Models

Multi-modal contrastive learning models [13, 7, 5, 6] represent a cutting-edge development in artificial intelligence. These models aim to bridge the gap between text and images (or other modalities) by learning powerful representations that connect words and pictures. The core idea is to train the model to understand the relationships between different modalities, enabling it to generate meaningful associations between textual descriptions and visual content.

One example of a multi-modal contrastive learning model is CLIP [13]. It leverages a massive dataset containing images and their associated textual descriptions from the internet. During training, it learns to encode both images and text into a shared

embedding space, where similar concepts from different modalities are closer to each other. This enables CLIP to perform a wide range of tasks, including image classification, object detection, and even natural language understanding, all from the same model.

One of the key advantages of multi-modal contrastive learning models is their versatility. They can be fine-tuned for specific tasks with relatively small amounts of labeled data, making them adaptable to a wide array of applications. For instance, CLIP can be used for zero-shot classification, where it can categorize images or understand textual descriptions of topics it has never seen during training.

Overall, multi-modal contrastive learning models represent a significant advancement in AI, enabling machines to understand and relate text and images in a way that mimics human-like comprehension. They have the potential to revolutionize various fields, from image recognition to natural language understanding, and their versatility makes them a powerful tool for tackling a wide range of complex tasks.

In the following chapters, we investigate multi-modal contrastive models and try to understand the embedding space resulting from the contrastive training. For an interactive version of this chapter visit <https://jku-vds-lab.at/amumo>.

2 Understanding and Comparing Multi-Modal Models

Contrastive Language Image Pre-training (CLIP) [13] and variations of this approach like, CyCLIP [7] or CLOOB [5], are trained on image-text pairs with a contrastive objective. The goal of contrastive loss objectives is to minimize latent-space distances of data points that have the same underlying meaning. We refer to the particular cases of contrastive learning that CLIP-like models perform as multi-modal contrastive learning because they use two (or more [6]) modes of data (e.g., images and texts) where each mode uses their own encoder to generate a latent embedding space. More specifically, the objective that CLIP is optimized for minimizes the distances between image-text embeddings of pairs that have the same semantic meaning while maximizing the distances to all other combinations of text and image embeddings. We would expect that such a shared latent space places similar concepts of images and texts close to each other, as demonstrated in Figure 2.1.

2.1 CLIP and its Modality Gap

Despite the clear objective that is supposed to bring texts and images to a shared embedding space there is a phenomenon called "Modality Gap" [9] which describes that embeddings of different modalities lie in their own embedding subspaces. The example in Figure 2.2 visualizes the Modality Gap between images and texts of CLIP¹ embeddings for a subset of 100 randomly selected images from MSCOCOs [10] validation set. The

¹We use the official CLIP implementation by OpenAI: <https://github.com/openai/CLIP> with the RN50 image encoder.

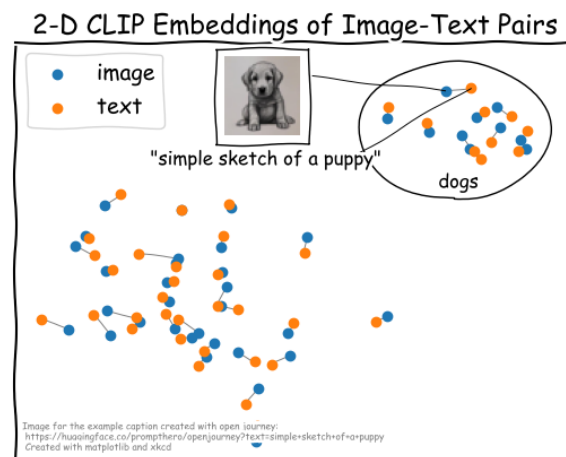


Fig. 2.1: A sketch of a projected bi-modal embedding space, as originally expected.

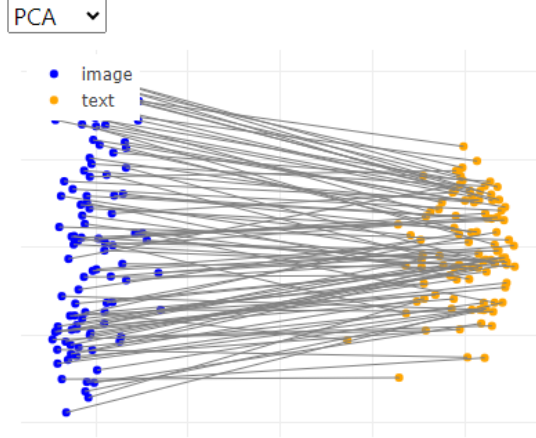


Fig. 2.2: Projected image and text embedding space as generated by CLIP [13].

image and text embeddings are projected to a 2-dimensional space and visualized in a scatter plot. We use a gray line to connect image-text pairs that belong together.

The use of dimensionality reduction methods to compute a 2-dimensional view of the data clearly shows the separation between the two modalities. However, dimensionality reduction comes hand in hand with a loss of information and possible distortion of data. We propose a different way to visualize the dimensionality gap. Similarity heatmaps are a simple yet effective way of visualizing latent space embeddings that also helped us to better understand the modality gap. Using this visualization also allowed us to gain interesting insights unrelated to the modality gap, which we could not have found with the scatter plot visualization alone.

The similarity heatmap in Figure 2.3 shows the same subset of 100 image-text pairs previously shown in a scatter plot. However, in this case, we show the cosine similarities calculated between all image and text embeddings. This results in a matrix with four quadrants:

- top-left: in-modal similarities of the 100 image embeddings,
- bottom-right: in-modal similarities of the 100 text embeddings,
- top-right: cross-modal similarities between the 100 image embeddings and the 100 text embeddings,
- bottom-left: a transposed version of the cross-modal image-text similarities.

The diagonal axis of each quadrant represents the matching data points (i.e., for the in-modal similarities, the diagonal shows the similarities of the image or text embedding to itself, while for the cross-modal similarities, the diagonal shows the matching image-text pairs). The modality gap is immediately visible: the in-modal similarities are much higher overall compared to the cross-modal similarities.

2.2 Where does the modality gap come from?

As analyzed by Liang et al. [9], the modality gap already appears before models are trained, possibly caused by random weight initialization and different model architec-

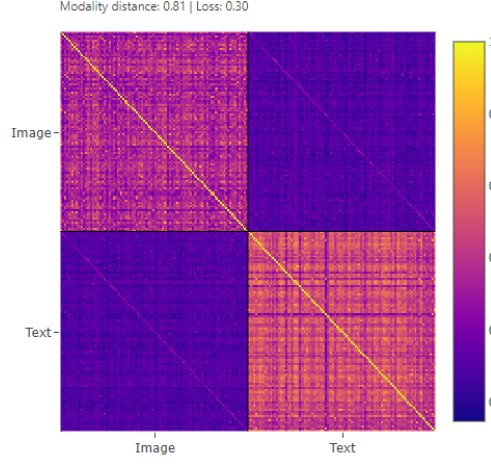


Fig. 2.3: Similarity of image and text embeddings visualized as heatmap.

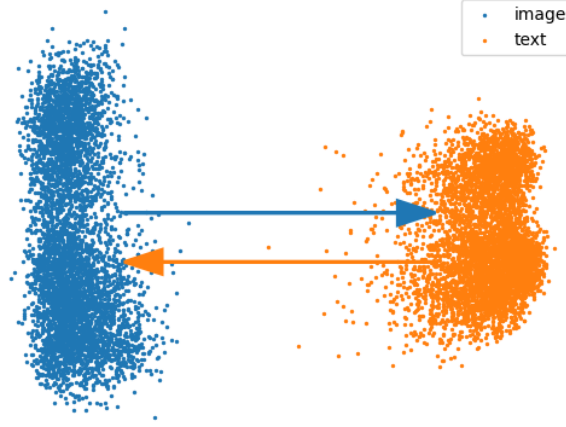


Fig. 2.4: Schema plot of how to close the modality gap by moving the embeddings along the gap vector.

tures. The authors also highlight that the gap between modalities persists throughout training, which means that CLIP’s objective function cannot overcome this phenomenon. A reason for that could be that the objective function only trains on the alignment of (non-)matching image-text combinations but does not contain any regularization terms with regard to the overall layout of the embedding spaces and in-modality alignment [7]. Liang et al. [9] also formally define the modality gap as the Euclidean difference between the centers of each modality:

$$\delta_{gap} = \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n y_i$$

where x_i and y_i are the normalized image and text embedding vectors.

They also experimented with manually reducing the modality gap by moving the embeddings closer together along the gap vector (see Figure 2.4).

However, since the modality subspaces trained by CLIP are not symmetric² to each

²Goel et al. [7] argue that the CLIP objective does, in fact, symmetrize the spaces in its optimal solution; in practice however this ideal scenario does not happen.

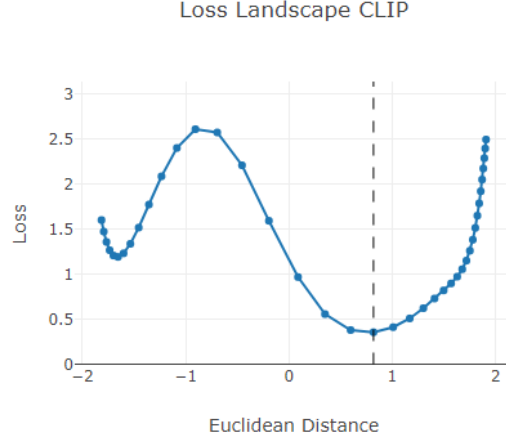


Fig. 2.5: Loss changes when moving the embeddings of two modalities in CLIP closer together or further apart. The dashed line marks the original distance of the two modalities.

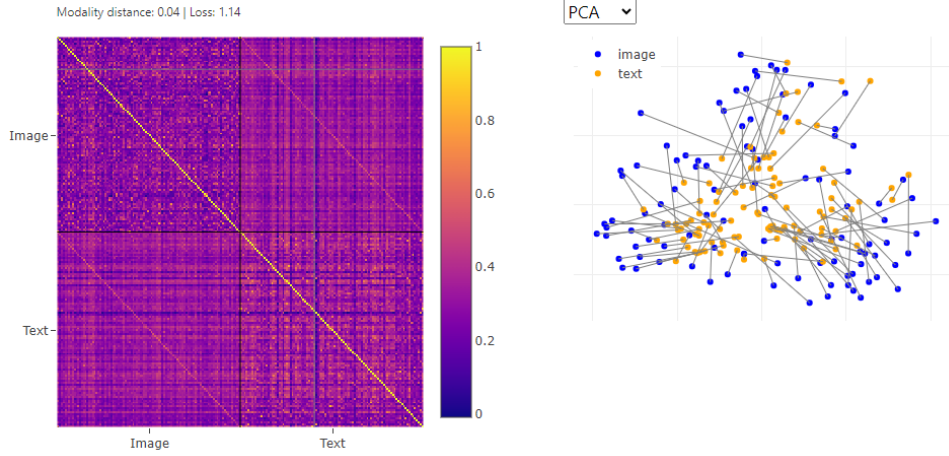


Fig. 2.6: Similarity heatmap and two-dimensional CLIP embeddings.

other, a modification of the embeddings destroys the complex relationship between images and texts that were derived during training. This modification naturally results in an increasing loss when changing the distance between modalities that were originally trained, as shown in the visualization in Figure 2.5. The x-axis shows the Euclidean distance between the two embedding centers (the black dashed line indicates the original distance). The y-axis shows the loss that results when moving the two embeddings closer together or further away from each other. We calculated these values with the whole 5000 samples MSCOCO validation set. It becomes visible that the global minimum of this manual intervention is at the point of the original (trained) modality gap.

Now, what do the similarity heatmap and scatter plot look like when we manually close the gap? As shown in Figure 2.6, the similarity matrix seems more homogeneous (i.e., in-modal similarities and cross-modal similarities are on a similar level), and points in the scatter plot are closer together. However, the scatter plot also shows that the edges between image-text pairs are still long. The many edge crossings between the

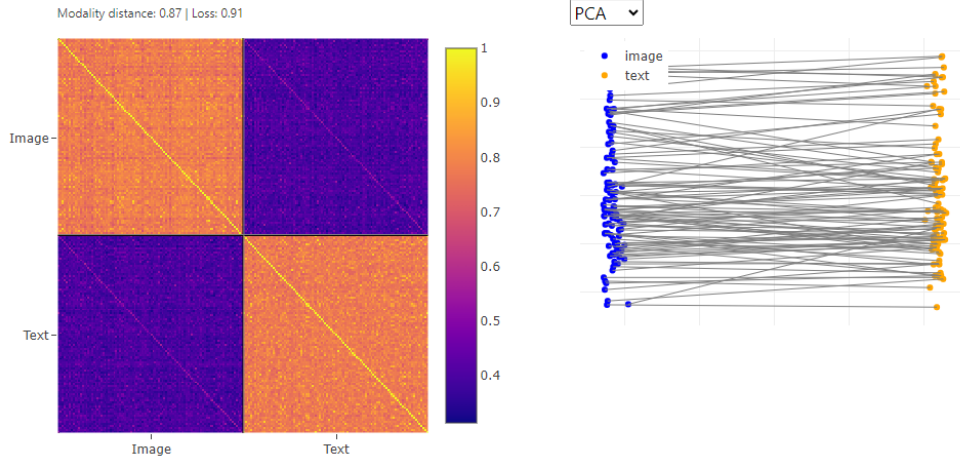


Fig. 2.7: Similarity heatmap and two-dimensional CLIP embeddings.

image-text pairs are also a qualitative indicator that the embeddings are not well aligned. Furthermore, the text embeddings concentrate more on the center compared to the image embeddings.

2.3 What About Other CLIP-Like Models?

As mentioned previously, the two modalities (i.e., texts and images) live on different embedding spaces, and the two embeddings vary in structure (i.e., they are not symmetric to each other). Recently published papers propose different versions of CLIP, where the objective function has been adjusted to regularize the trained embedding space. We look closer into two promising approaches, namely, CyCLIP [7] and CLOOB [5].

2.3.1 CyCLIP

According to Goel et al. [7] using CLIP-generated image-text embeddings interchangeably (e.g., for Language-guided image generation) is suboptimal because the embeddings are—in practice—not aligned. They propose an augmentation of the InfoNCE loss used to optimize CLIP that adds two regularization terms that enforce the two embedding spaces to be symmetric: one for in-modal symmetry (LI) and one for cross-modal symmetry (LC) of the similarities.

$$L_{CyCLIP} = L_{InfoNCE} + L_I + L_C$$

The visualizations shown in Figure 2.7 already hint that the embeddings are symmetric: The in-modal similarity heatmaps of the image and text quadrants look similar, and the modalities in the image-text points in the PCA projection seem almost parallel. However, it is also clear that the modality gap is still present.

We can confirm this with a slight modification of the similarity heatmap. For this, we calculate the difference between the in-modal similarities of images and texts, which results in a matrix where values approach zero (i.e., the matrix is a blue rectangle).

When looking back to Liang et al.’s [9] experiments that tried to align the embeddings by moving them along the modality gap vector, we saw that this does not work

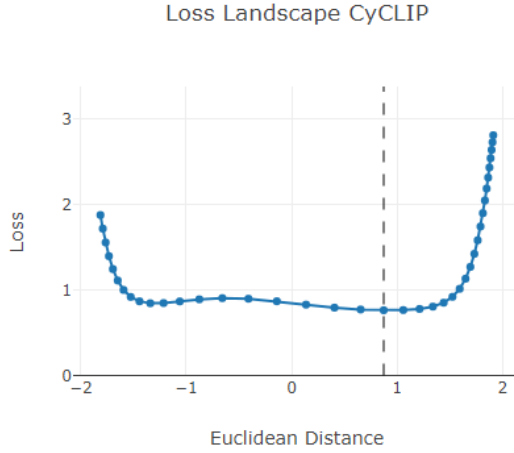


Fig. 2.8: Loss changes when moving the embeddings of two modalities in CyCLIP closer together or further apart. The dashed line marks the original distance of the two modalities.

well for CLIP embeddings because the space is not symmetrized. For CyCLIP, however, the symmetrization is enforced with the objective function, and we could see in the previous plots that the spaces are indeed mostly symmetric. Let’s look into the loss landscape for different modality distances of CyCLIP embeddings when moving the embeddings along the gap vector. In Figure 2.8, we can see that CyCLIP’s loss landscape looks very different from CLIP’s landscape: in the distance interval of $[-1.5; 1.5]$ the loss does not change much, which is another indicator that CyCLIP indeed learns nearly symmetric embedding spaces³.

Again, when closing the modality gap, the similarity heatmap becomes more homogenous (i.e., the in-modal similarities and cross-modal similarities are similarly strong). While the image-text pairs in the scatter plot are closer, there are still quite some edge crossings (see Figure 2.9).

Alternatively, we can use UMAP or tSNE projection to better utilize the neighborhoods of similar embeddings instead of linearly determining the axes with the highest variance, as done in PCA (see Figure 2.10). In the case of CyCLIP (scatter plot on the right hand), the use of a neighborhood-based dimensionality reduction technique results in shorter edges and better clustering of similar embeddings. For CLIP (scatter plot on the left hand), edges remain long, and similar images and texts are not clustered together.

Let us summarize how the loss changes for CLIP and CyCLIP when moving the embeddings together along the modality gap vector. For that, we again use the entire MSCOCO validation dataset of 5000 samples.

Model	Orig. Dist.	Orig. Loss	Closed Dist.	Closed Loss	Loss Diff.
CLIP	0.818611	0.355370	0.035077	1.124780	0.769410
CyCLIP	0.873026	0.763433	0.001218	0.848867	0.085434

The numbers confirm that CyCLIP’s loss changes far less than CLIP’s loss when

³Note that the loss outside the $[-1.5; 1.5]$ interval increases. This could be due to the fact that the spaces are not perfectly symmetric. Another explanation could be that this happens due to normalizing the embeddings to the unit sphere, but the method proposed for closing the gap is done in Euclidean space.

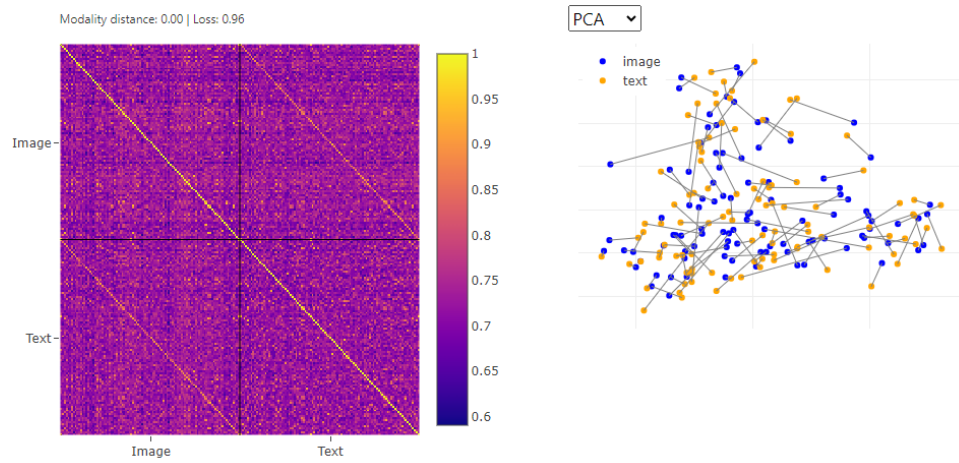


Fig. 2.9: Similarity heatmap and two-dimensional CyCLIP embeddings.

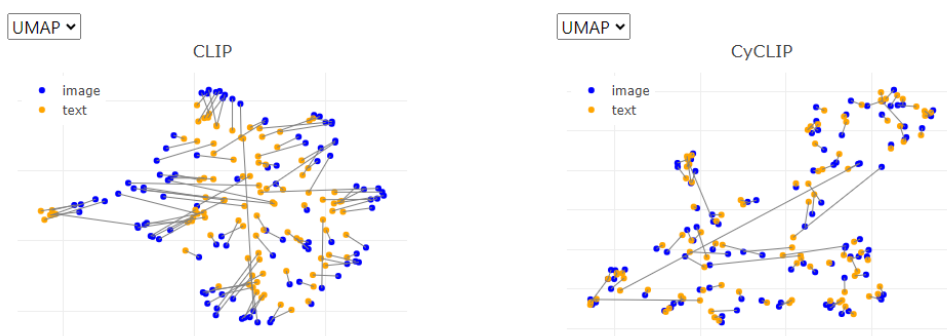


Fig. 2.10: Two-dimensional CLIP embeddings projected with UMAP.

manually closing the gap – it neither gets better nor significantly worse. However, the question arises: Why Do We Even Want to Close the Modality Gap, if We Do Not Gain Performance?

2.3.2 Why Do We Even Want to Close the Modality Gap?

From a performance optimization point of view, this is a valid question. What's the point of interfering in a well-performing system? On the other hand, the alignment of embedding spaces can become important for other downstream tasks. For example, using image and text embeddings interchangeably, as done in language-guided image generation, relies on the fact that image and text embeddings are aligned with each other and live in the same space. Another aspect is that an aligned embedding space is closer to how humans expect multi-modal models to see the data. Furthermore, closing the modality gap allows us to actually visualize texts and images in the same space, and develop interactive exploration tools that help to understand multi-modal data (e.g., analyzing pairs of human written captions and machine-generated images to find insights about text-to-image generation models like StableDiffusion [17]).

These example use cases of why closing the modality gap might be helpful should give you an incentive about why the pure "performance optimization" point of view is not the only one. In fact, if we can close the modality gap without significantly losing performance, we have a win-win situation!

2.3.3 CLOOB

In the previous section, we established a way to manually close the modality gap of CyCLIP embeddings. However, wouldn't it be better to already close the gap during training and not rely on post-hoc manipulations? While we did not experiment with further modifying (Cy)CLIP's objective to close the gap during training, we stumbled upon a different learning approach that naturally closes the gap.

Contrastive Leave One Out Boost (short: CLOOB) [5] is a variation of CLIP that proposes an alternative objective together with an associative memory to train the model. The two main components of their method are (i) modern Hopfield networks and (ii) the InfoLOOB loss instead of the InfoNCE loss used by CLIP. The authors argue that their modifications solve CLIP's "explaining away" problem (i.e., focusing on a small subset of features while ignoring other relevant features) and InfoNCE's saturation problem⁴.

However, we also observe something else: These modifications seem to aid the closure of the modality gap, as shown in Figure 2.11.

We believe that CLOOB's ability to close the gap during training mainly stems from the cross-modality retrieval applied before calculating the InfoLOOB loss. In this step, the batch of image embeddings and text embeddings is used as an associative memory to create a weighted average of embeddings for the current instance. Since this is done for each combination of embedding and associative memory (i.e., each image and each text embedding is associated with the entire batch of image embeddings AND the entire batch of text embeddings), a stronger correlation between the image/text instances to their own modality and the opposite modality is established. Other influencing

⁴See Rumetshofer et al. [5] for more information.

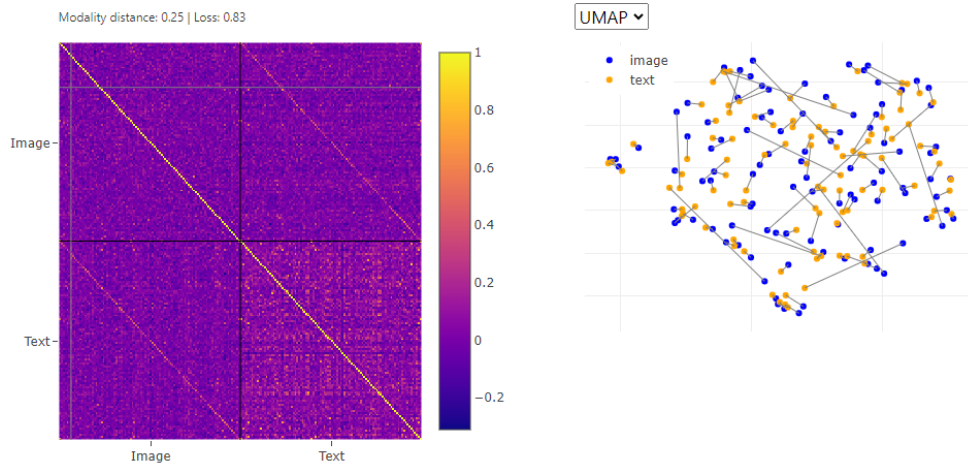


Fig. 2.11: Similarity heatmap and two-dimensional CLOOB embeddings.

factors on closing the modality gap could be the use of InfoLOOB, or the use of modern Hopfield networks that generally have a denoising effect⁵.

2.4 Closing the Gap Using a Larger Dataset

Amumo - our interactive prototype - is an easy way to explore a small subset of image-text pairs. This analysis can help form intuition about a particular dataset or the model used to map them into a latent space embedding. In addition to the interactive prototype, we also want to showcase the results of the proposed methods for closing the modality gap with a larger dataset. To that end, we again used the entire 5000 sample MSCOCO validation dataset and applied the two methods introduced in the article. We then take the aligned embeddings, project them with UMAP, and plot them in a static 2-d scatter plot with lines connecting the matching image-text pairs.

2.4.1 Manually Removing the Modality Gap

For the first method to remove the modality gap, we need to compute CyCLIP embeddings and manually move the two embedding spaces together. The scatter plot in Figure 2.12 shows the 2-dimensional projection of these modified embeddings. The plot shows that clusters are forming and a lot of connection lines are within clusters, which means that instances that carry a similar meaning are indeed close together in the latent space embedding. However, there are also a lot of intra-cluster connections. The emergence of these long connections between the clusters can be caused by various factors. For example, the manual modification of the latent space might disturb some parts of the latent space. Although CyCLIP does add restrictions to the objective function to facilitate the emergence of symmetric embedding spaces, it is only an approximation and the spaces are not perfectly perpendicular. Another influencing factor might be that the image-text pairs are not perceived as similar by the model and therefore placed in different areas of the embedding space. Finally, there may also be distortions coming

⁵See this blog post for a detailed explanation of modern hopfield networks: <https://ml-jku.github.io/hopfield-layers/>.

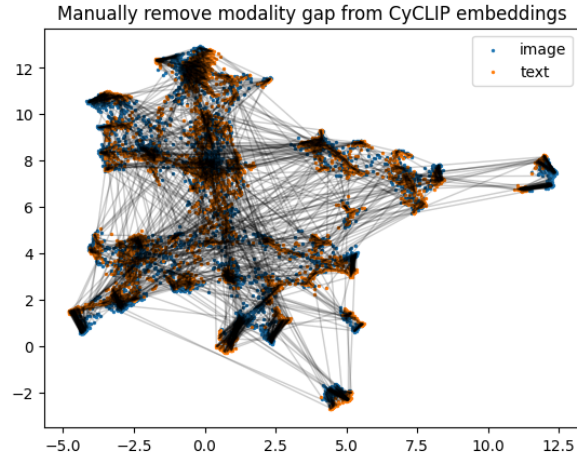


Fig. 2.12: Two-dimensional projection of CyCLIP embeddings with the modality gap manually removed.

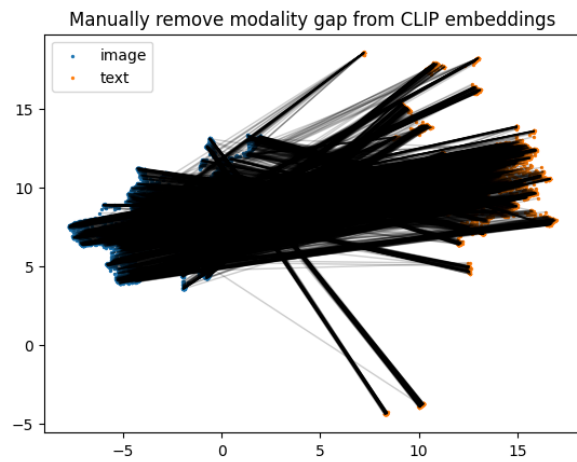


Fig. 2.13: Two-dimensional projection of CLIP embeddings with the modality gap manually removed.

from the dimensionality reduction technique we used for creating the 2-d space. To investigate these assumptions further, we recommend the use of interactive tools that allow exploring large sets of points and clusters in a 2-d space (e.g., the Projection Space Explorer [3]).

The plot in Figure 2.13 shows the same procedure of manually removing the modality gap, but here we used CLIP embeddings instead. This shows again, that manually removing the gap by moving CLIP embeddings on the same plane destroys the trained latent space too much to be useful anymore.

2.4.2 Inherently Removing the Modality Gap

The second method of removing the modality gap utilizes a bi-modal model capable of aligning the embedding spaces: CLOOB. In this case, we can directly use the image and text embeddings generated by the model and project it to a 2-d space for visualization. When looking at Figure 2.14, we see a similar result to before: clusters emerge and a lot

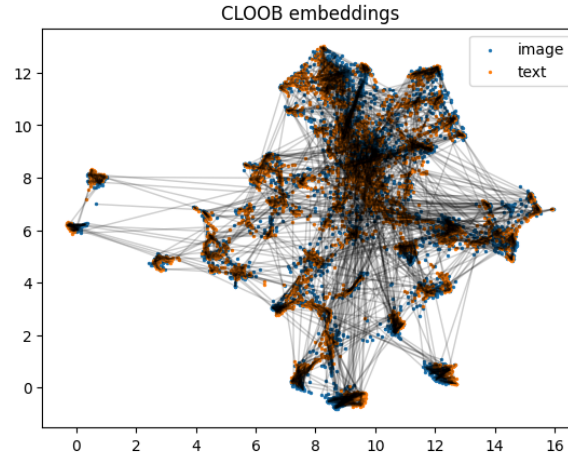


Fig. 2.14: Two-dimensional projection of CLOOB embeddings with the modality gap inherently removed.

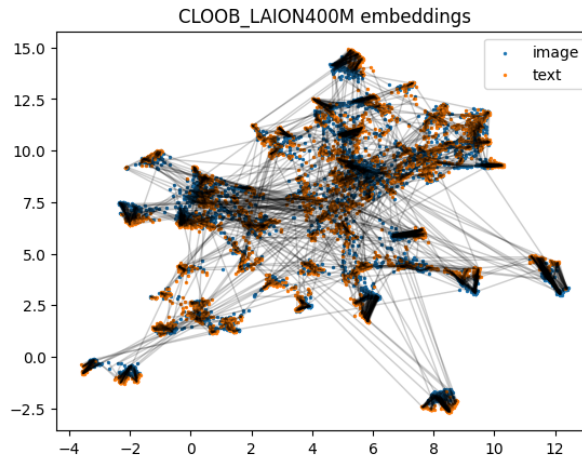


Fig. 2.15: Two-dimensional projection of CLOOB Laion400M embeddings with the modality gap inherently removed.

of inter-cluster connections, but also plenty of intra-cluster connections. There also seems to be a rather large cluster of points with many connections.

In comparison, we also show the results for the same model, but trained with 400M instances of the LAION dataset and a vision transformer architecture used for image embedding instead of a CNN architecture. In Figure 2.15, we can see smaller cluster entities and it seems like connections between clusters are less. To confirm these qualitative findings and be able to compare the methods, we would have to use quantitative measure (e.g., measuring how many intra-cluster connections there are for each method).

2.5 Visually Closing the Modality Gap

In the previous sections, we learned about two ways that can help us close the modality gap:

1. Manually: post-process embeddings by moving them together along the modality gap vector; this only makes sense if the two embedding spaces are symmetric, like in CyCLIP.
2. Inherently: define the model architecture and/or training objective in a way that aids the closing of the gap, like for CLOOB.

Let's also recall the reasons why we would like to close the gap:

1. can have advantages for downstream tasks (e.g., if embeddings from two modalities need to be interchangeable)
2. aids the development of multi-modal visual analytics tools
3. match human expectations of how the embedding space should look like

From a visualization point of view, we might not care about the other two reasons, as long as we can visually close the modality gap. By visually closing the gap, we do not change the embeddings or the model, but map them into a shared low-dimensional space. This can be accomplished in several ways: (i) out-of-sample projection, (ii) concatenating image-text embeddings and treating them as one combined embedding. The first method results in a low-dimensional datapoint for each image and each text embedding; the second method results in a combined low-dimensional space for the image-text pairs. The method you would want to choose depends on the goal of the visualization.

2.5.1 Out-of-Sample Projection

We can first project embeddings of either images or texts using UMAP. This projection builds a neighborhood graph using in-modal similarities and results in a low-dimensional projection for one modality. We can utilize UMAP's out-of-sample projection to also project the embeddings of the second modality onto the space of the first modality. Since the out-of-sample projection again tries to map each point to the most similar points in the existing low-dimensional space, you can imagine this as using the cross-modal similarities between images and texts. Since this is what CLIP was trained on (i.e., optimizing distances between texts and images), the mapping should visually remove the modality gap.

As an example, we take the 5000 sample MSCOCO validation set and first fit and transform the image embeddings. We then transform the text embeddings with the existing UMAP embedding and show the results in a scatter plot. As shown in Figure 2.16, the overall structure seems to align images and texts; while there are a lot of cross-cluster connections that show that image-text pairs are not always close to each other, most of the connections seem to be within clusters. Cross-cluster connections can be indicators of various things. For example, the pairs may not be deemed similar by CLIP, which results in embeddings that are far away from each other in the high-dimensional latent space, which would be reflected in the low-dimensional projection of the embeddings. Another issue might come from the projection method itself. As mentioned previously, projecting data to a low-dimensional space comes with a loss of information that might introduce artifacts. The fact that we use out-of-sample projection might amplify this effect even further.

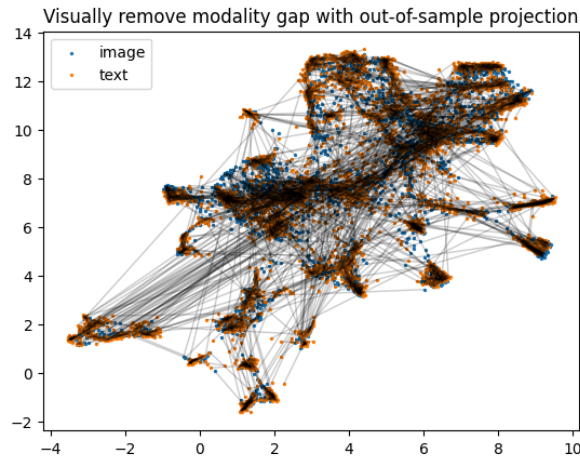


Fig. 2.16: Two-dimensional projection of CLIP embeddings with the modality gap removed using out-of-sample projection of the text embeddings.

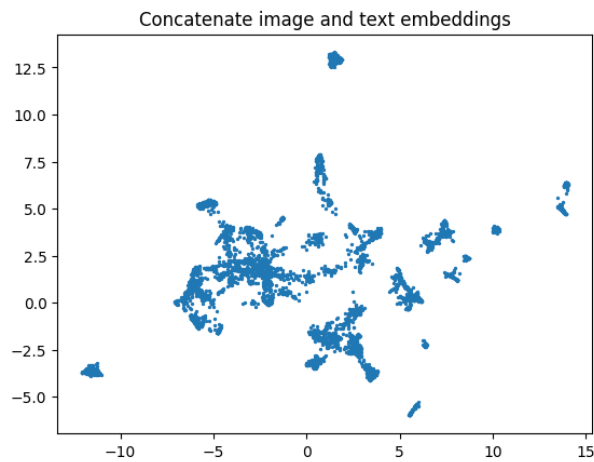


Fig. 2.17: Two-dimensional projection of CLIP embeddings where image and text embeddings were concatenated to one datapoint.

Visual analytics tools can be helpful in gaining further insights into the data and why certain pairs seem to be far away from each other. Basic interactions like hover information or selection summaries could already be a good start for further investigation. The rich nature of image and text data also allows for more advanced analytic visualizations; for example, texts can be used to extract labels for clusters that carry rich semantic meaning, or example images could be used to summarize clusters. Visually encoding the high-dimensional similarity of embedding pairs (e.g., as saturation of the lines between pairs) could be a helpful indicator that could show whether point pairs are far apart from each other due to artifacts from the projection or due to CLIP not recognizing them to be similar.

2.5.2 Concatenating Image and Text Embeddings

For a different kind of visual representation of image-text embedding spaces, we can simply concatenate the embedding vectors and transform them into a combined low-

dimensional space. The low-dimensional space can be visualized in a scatter plot and visual analytics approaches can be used to explore the data. Note that with this approach, we only have one low-dimensional data point per image-text pair as shown in Figure 2.17.

2.6 Summary of Modality Gap Analysis

The visualizations in Figure 2.18 give an overview of the similarity heatmap for CLIP and CyCLIP before (left) and after (right) manually removing the modality gap, as well as the similarity heatmap for CLOOB with the official checkpoints from the paper and a CLOOB version that was trained on the LAION 400M dataset and used a ViT instead of a CNN to encode images [2]. Note how the overall distribution of similarity values differs between the various embedding spaces. For example, the two CLOOB models seem to discriminate more strictly between matching and non-matching image-text pairs.

The previous sections taught us about the modality gap, where it comes from, and ways to close it. We also gave reasons for why closing the gap might be beneficial and now demonstrate how closing the modality gap can help with analyzing multi-modal data. We also introduced a handy new way of visualizing latent space embeddings and utilize this again in the following analyses. Finally, we also want to mention that there is a way to visually close the modality gap (i.e., without actually closing the gap in the embedding space).

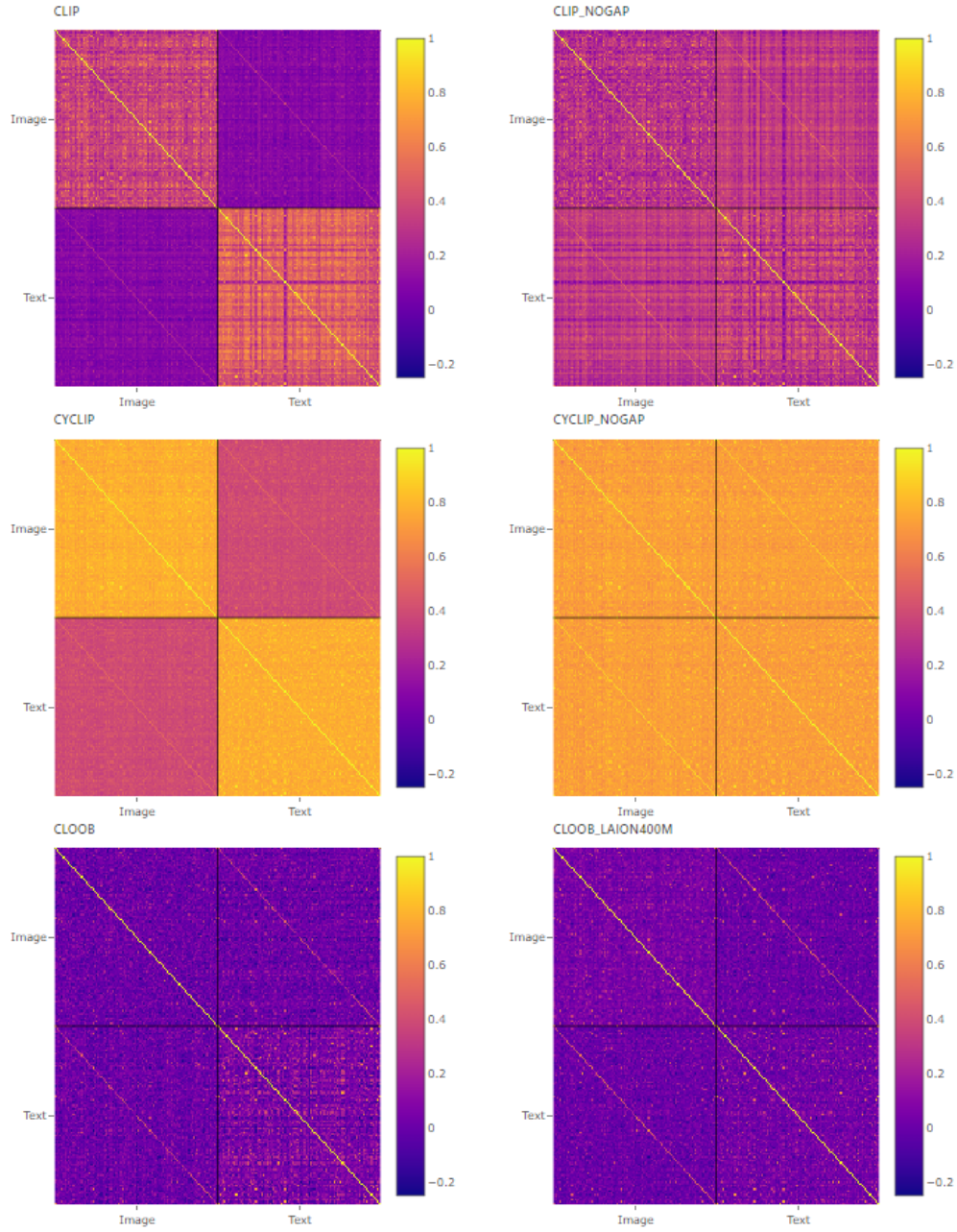


Fig. 2.18: Similarity heatmaps for all CLIP-like models and their closed embeddings.

3 Converting the Technique into a Tool

Using the previously introduced techniques, we implemented an interactive prototype called “Amumo” (Analyze Multi-Modal Models). Users can switch between models, explore the similarity heatmap and scatter plot visualizations, manually close the modality gap, and try various projection methods (see Figure 3.1).

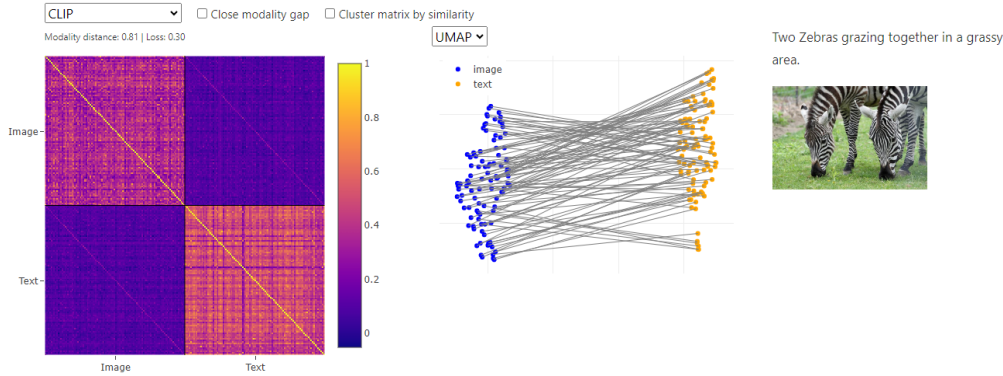


Fig. 3.1: Interactive tool to analyze multi-modal models "Amumo".

3.1 Identifying Data Subsets

We can look into semantic subsets of data by filtering instances based on their captions. The following example shows the visualizations for the subset that contains the substring "dog". We notice that some lines in the similarity matrix have a darker color. When hovering over those darker lines, we can see that most of these instances correspond to images and texts about "hot dogs" or other images that do not show a dog or where a dog is in an uncommon setting. To make this even more obvious, we can use the "Cluster matrix by similarity" function that reorders the similarity heatmap such that similar lines are grouped together. One cluster that stands out in all three CLIP-like models is the "hot dog" cluster. However, we can also see clusters for "dog and frisbee", "dog and bed", or "dog and car". A screenshot of the analysis is shown in Figure 3.2.

3.2 Analyze DiffusionDB Dataset

We would like to see what the models' latent-space embeddings look like for a dataset that is not (entirely) procured by humans. To this end, we use DiffusionDB [21], a collection of human-written captions and images generated from these captions by Stable Diffusion [17]. We use a subset of 100 randomly selected samples to qualitatively explore the embedding spaces created by the CLIP models we previously introduced. An example of the analysis is shown in Figure 3.3.

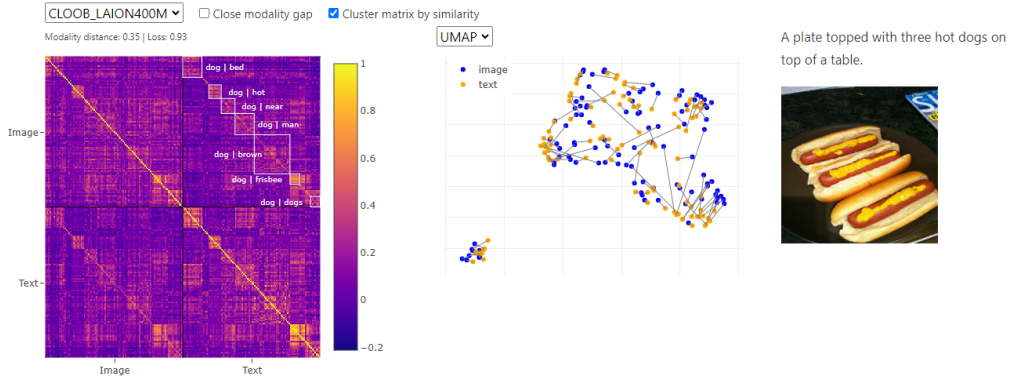


Fig. 3.2: Using Amumo to analyze a data subset of "dog". Reordering the similarity heatmap reveals a cluster that shows hot dogs instead of the animal.

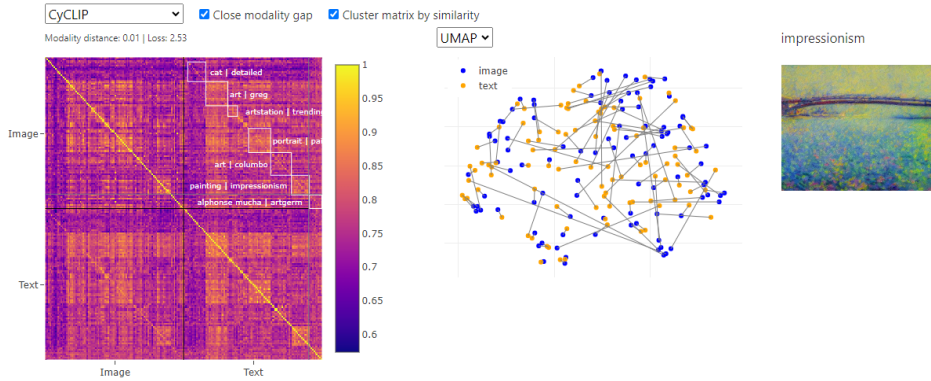


Fig. 3.3: Using Amumo to analyze the DiffusionDB dataset.

With the default settings, we randomly explore the dataset and get a feeling for the data contained in this subset. We can investigate instances that are outliers in the similarity heatmap by hovering rows or cells that have particularly large or low similarity values. For example, there are some particularly bright cells scattered in the image in-modal similarity heatmap. Upon hovering, we see that all of these images are blurry. We know that DiffusionDB added blur filters for images that were detected to show inappropriate content. Interestingly, CLIP seems to create similar latent embeddings for blurry items, causing them to show high similarity in the similarity heatmap.

For further analyses, we choose "Cluster matrix by similarity" to order the matrix in a way that groups similar rows in the heatmap and investigate the clusters that are emerging. We can see a cluster for "impressionism and crystal" that seems to have homomorphic similarities over all images (i.e., there is a distinct purple line along all images of this cluster). Upon further investigation, we see that the captions in this cluster are mostly vague texts or single words (e.g., "crystal", "impressionism") that can apply to a lot of images. The same cluster becomes apparent in the text in-modal similarity heatmap, where all captions within the cluster seem to have high similarity.

Let's close the modality gap to investigate clusters in a 2-dimensional scatter plot. We can either do this by switching to the CLOOB model or using CyCLIP in combination with the "Close modality gap" option. We see that the embeddings are aligned and can use the interactive scatter plot to investigate clusters. For example, we can try to

find the cluster of blurry images, or we can try to find the cluster with instances of "impressionism". Of course, this would be much more fun on a larger scale :)

3.3 Augmentation Analyses

As previously demonstrated, we can identify patterns in datasets and subsets of datasets using the similarity heatmap visualizations. Now, we would also like to see if we can use the same techniques to find patterns in augmentations of a single data point. For example, we take a single image, generate rotated versions of this image, and use this augmented dataset to compute CLIP embeddings and similarities. The results of this experiment for the three CLIP-like models are shown in the visualization in Figure 3.4 (note that we again show two variants of the CLOOB model). To generate this dataset, we gradually rotate a selected image by 360 degrees over the course of 100 steps. Each step results in a “new” image and a new data point. Note that we only augment the image, but not the text, which results in a completely homogeneous similarity in the in-modal text quadrant of the heatmap and homogenous stripes along the text dimension in the cross-modal quadrants of the heatmap.

When looking at the in-modal image similarity quadrant of the heatmap for each model using augmentations of the first image, we can see an interesting pattern emerge. In addition to the bright yellow diagonal axis that corresponds to the similarities of images to themselves, there is also the perpendicular off-diagonal axis of the matrix sticking out. When hovering along the off-diagonal, we see that the two images along this axis are actually mirrored versions of each other. It seems like all models are invariant to the horizontal flip transformation for this image. We can also see a checkerboard-like pattern emerge for some images emerging in all models except for the CLOOB_LAION400M. When looking into the darker areas of this heatmap in more detail, we can see that the pattern occurs around multiples of 90-degree rotations. The fact that this pattern occurs mainly for the three models that use a CNN-based image encoder¹ and not for the one with the vision transformer could be an indicator that the two architectures vary in their ability to learn rotation invariant properties. The checkerboard-like pattern seems to be consistent with findings described by Timme et al. [19] where they tested the rotation robustness of various CNN classifiers by measuring the accuracy. The accuracy of the CNNs showed local maxima at multiples of 90-degree rotations and was lower in-between those angles.

When looking at the overall distribution of similarity values, we also notice that the ViT-based CLOOB model seems to have more patches of low-similarity values compared to its CNN-based counterparts. This might indicate that ViT’s overall robustness to rotation transformations is lower. In further investigations, we might want to directly compare two versions of CLIP: the current version with the CNN-based image encoder and a version with a ViT-based image encoder, and study the phenomenon on a larger dataset.

Use the interactions in the online article to explore the heatmaps for different images yourself.

In a second experiment, we analyze the heatmaps for an image to which we add an

¹Note that CLIP and CLOOB_LAION400M (both 400M instances) were trained on a much larger dataset than CLOOB and CyCLIP, which could also be an indicator for varying robustness.

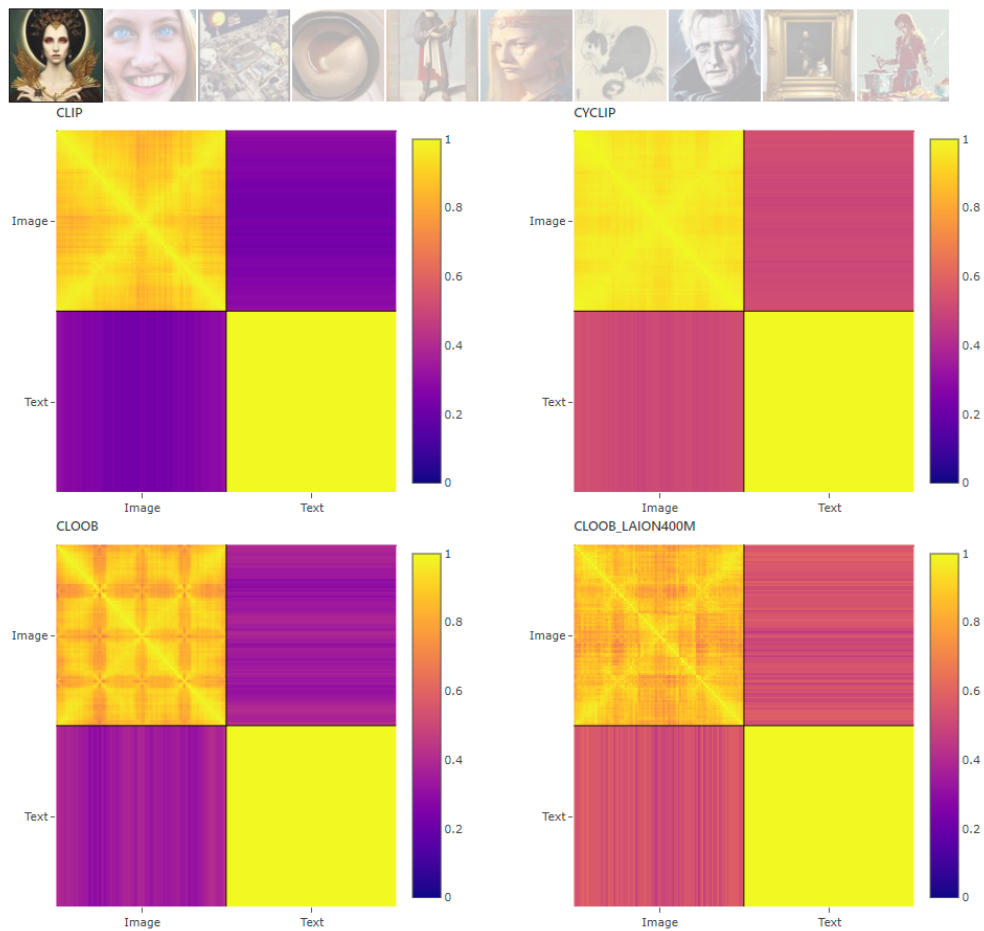


Fig. 3.4: Using Amumo to analyze embeddings of rotated versions of an image.

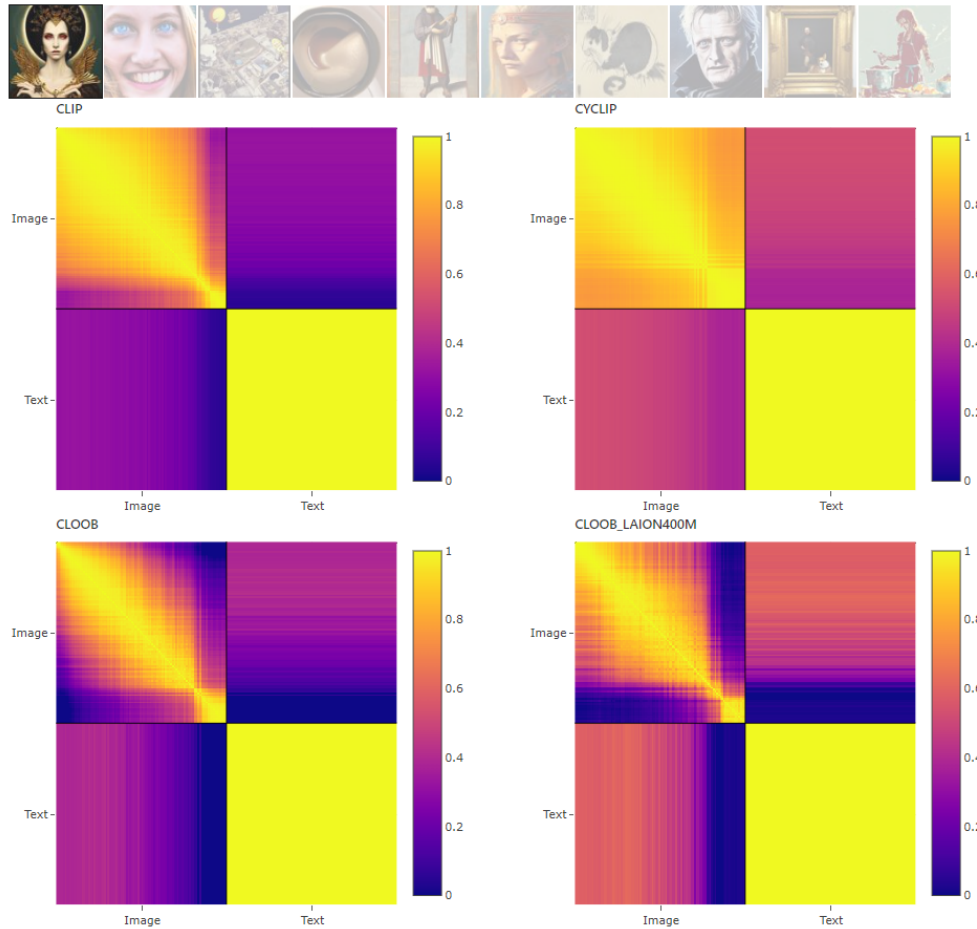


Fig. 3.5: Using Amumo to analyze embeddings of noisy versions of an image.

increasingly higher noise level. When looking at the heatmaps for the first image in Figure 3.5, it seems like there is a certain level of noise for each model, after which the model cannot seem to recognize the content of the image anymore. All images with a higher level of noise than this threshold seem to look (almost) the same (as indicated by a bright yellow rectangle at the lower-right corner of the in-modal image similarity quadrant).

Similarly, for blurry images, we see in Figure 3.6 that at a certain point of blurriness, all images look the same to the models, and they cannot map images and texts together. You can use the dropdown menu in the online article to explore the effects of various augmentations.

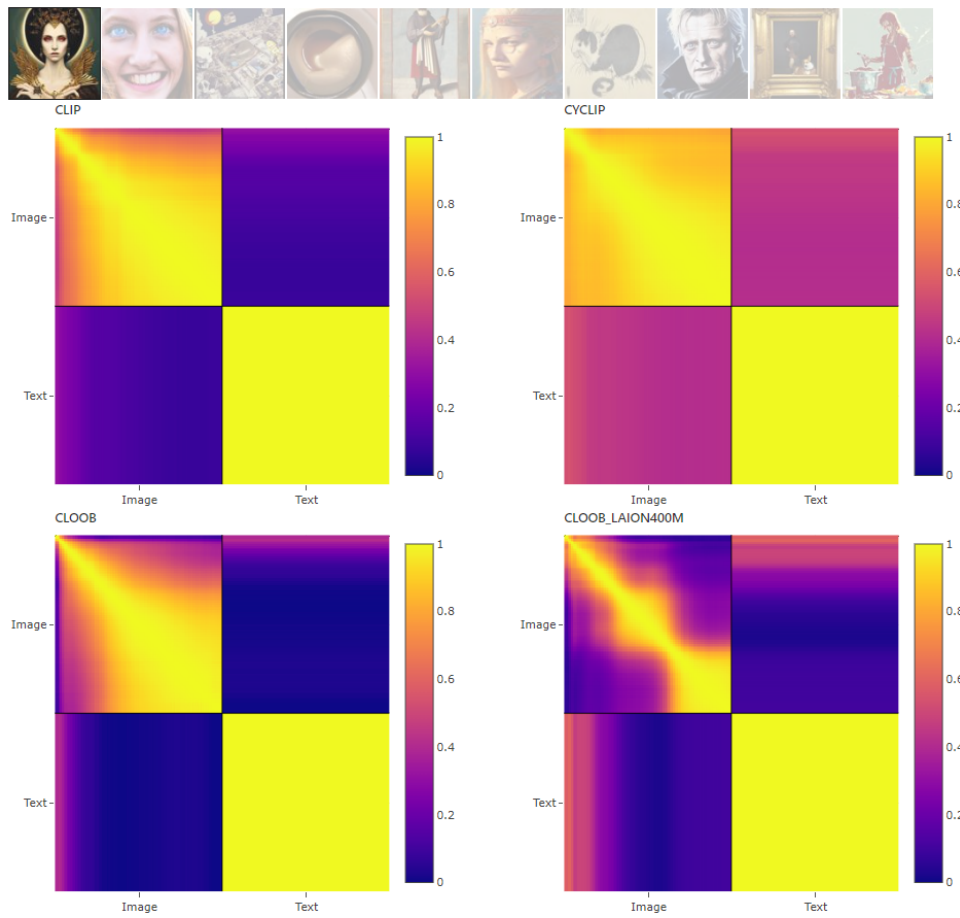


Fig. 3.6: Using Amumo to analyze embeddings of blurry versions of an image.

4 Conclusion

Throughout this article, we investigated latent embeddings of CLIP-like models. Using scatter plots and similarity heatmaps, we visualized and analyzed the modality gap that naturally occurs for CLIP embeddings. Closing this gap without losing significant performance can be important for downstream tasks like image generation, visual analytics, or human understanding. We showed how to close the gap using CyCLIP in combination with a post-processing method that aligns the embedding spaces and investigated another model (CLOOB) that is able to align the spaces during training. Finally, we introduced Amumo, an interactive visual prototype that allows users to explore embeddings from bi-modal contrastive learning models to help with understanding of their latent space embeddings. We used Amumo to analyze various (sub-)sets and augmentations of data. We believe that Amumo, and the similarity heatmap in particular, are useful tools to create intuition about bi-modal latent space embeddings. It allows for comparison of bi-modal models (e.g., their robustness to transformations) and can help to formulate hypotheses or ideas about such models. However, we want to stress that the analysis is based on a small subset of data points, and insights must still be verified on a larger scale.

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